Natural Selection

Organisms with characteristics that help them to survive in their environment have a greater chance of reproducing and producing young. Their young will inherit these characteristics. As a result the number of organisms with these favorable characteristics increases.

Here is a famous example. Below are two pictures of types of peppered moths.



1. Birds like to eat these moths. If all the trees in the environment have black bark, which moth will have the most chance of surviving? Bark is the outer "skin" of the tree.

2. What is the characteristic this moth has that helps it survive?

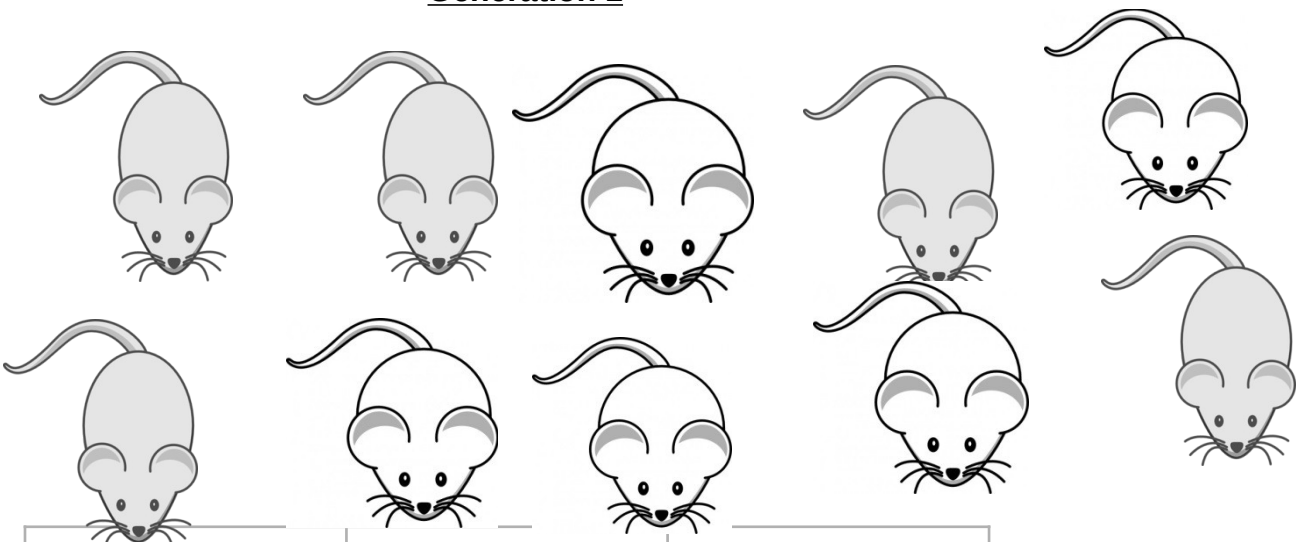
3. After 10 years which moth do you think there will be the most of?

4. Use your answers to fill in the sentences that follow.

The peppered moth comes in two c_____. Two hundred years ago pollution turned the bark of trees where the moths lived black. As a result the black moths had a characteristic that helped them to survive in their e_____. They had a greater chance of reproducing and producing young. So over time the number of black moths increased and the number of white moths _____.

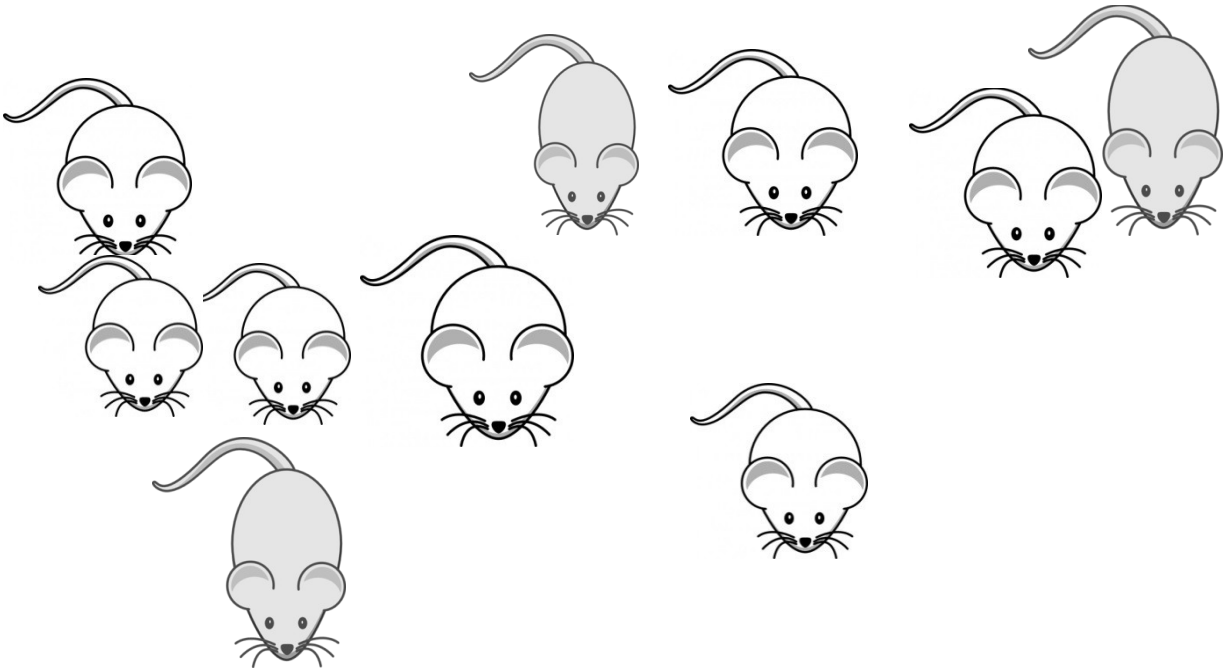
The mice shown below live on the sand dunes near a beach. The sand is white. The mice are hunted by birds called hawks. All the mice are in the same species. Some have dark fur and some have light fur. The numbers of the different coloured mice were counted over 3 generations. Fill in the tables for the three generations and complete the questions at the end.

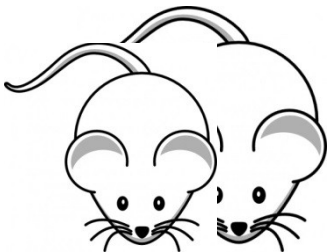
Generation 1



Generation 1	Dark	White
number of mice		

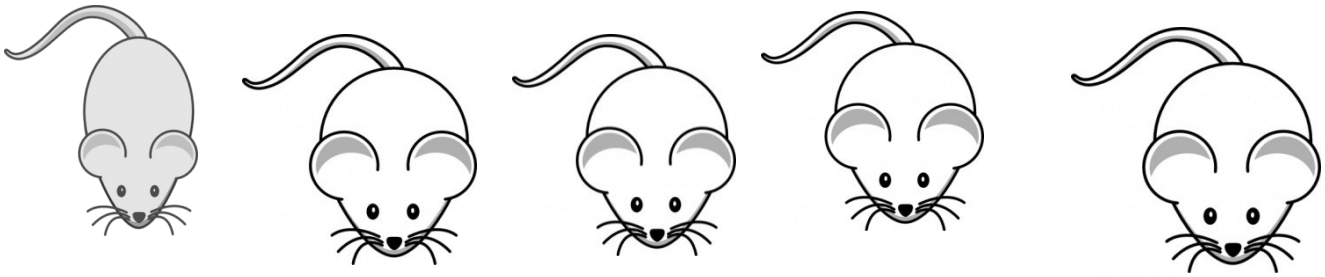
Generation2.



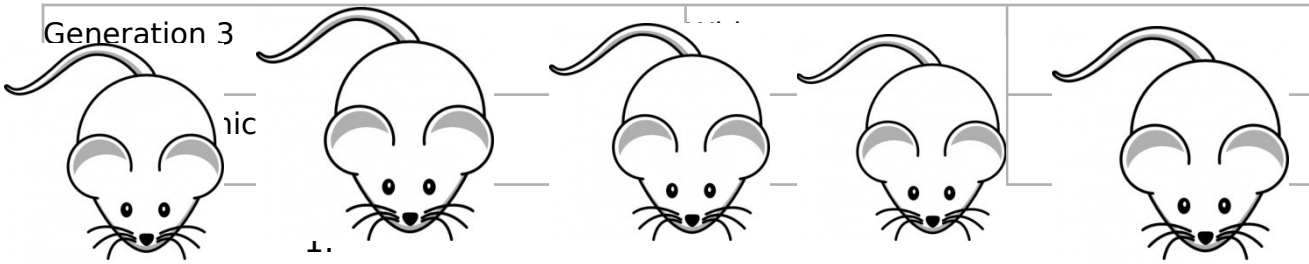


number of mice			

Generation 3.



Generation 3			

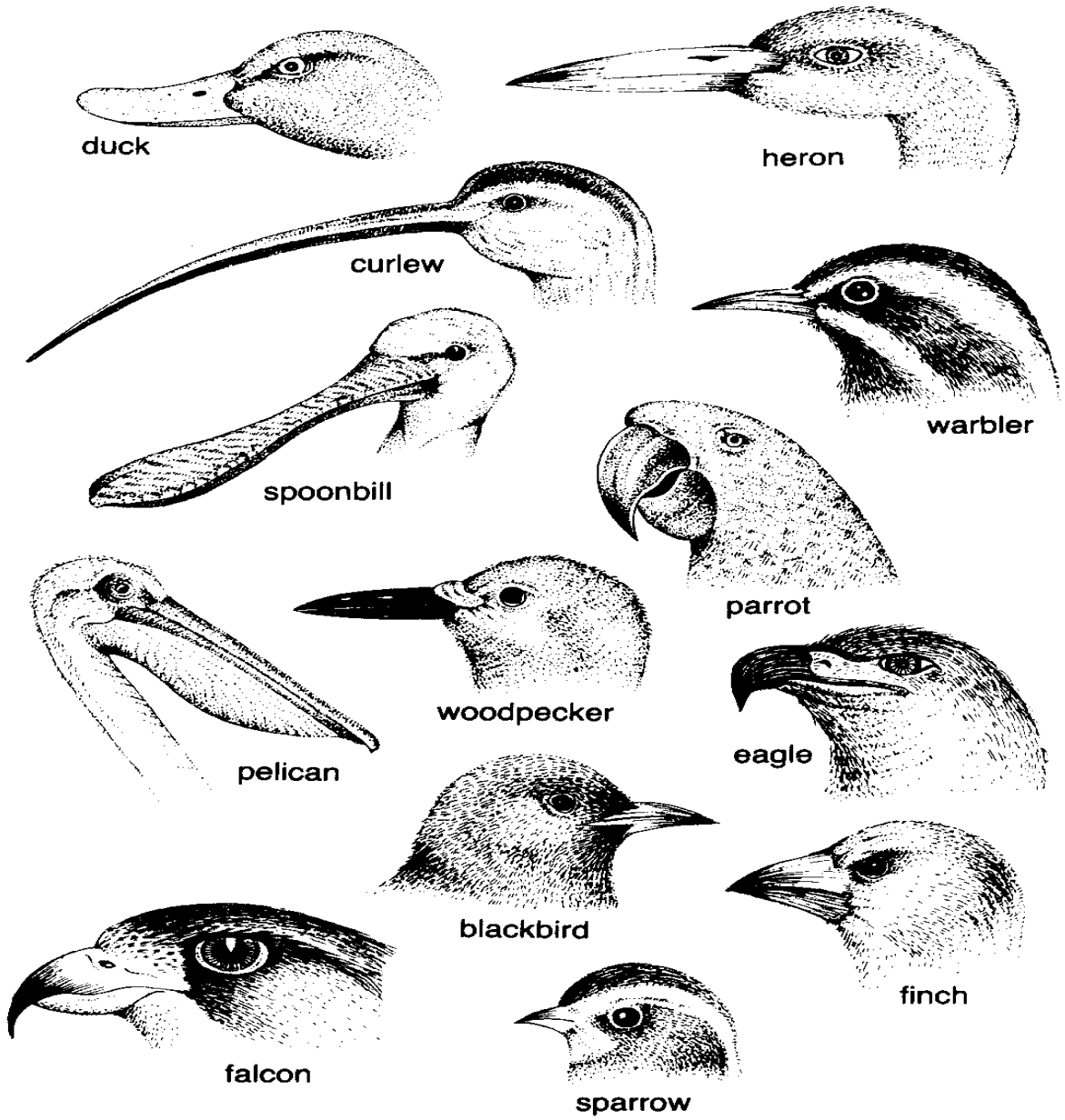


Which colour is being selected for?

2. Why do you think this colour is being selected for?

We call this colour that is being selected for a FAVOURABLE CHARACTERISTIC.
Favourable characteristics increase an organism's chances of survival. Natural selection increases the number of organisms with favourable characteristics.

Look at these crazy birds. They have beaks that are favourable characteristics that help them get food. Here are some ways they get food: digging for worms and shells in mud, ripping meat of animals that they have killed, catching fish, cracking open nuts, picking up small seeds and insects. Beside the picture of the birds write how you think they get their food.



Biodiversity

Biodiversity means that there are many different s_____.
A rain forest has a lot of different species, so it has high b_____.
A field of corn plants only contains a few species so it has low b_____.
The process of **speciation** is the process which produces all of these new species.

Speciation

This is the process where new species are produced. It happens in a number of steps

1. Variation

2. Isolation

3. Selection

4. New Species

Questions

1. Dogs are all members of the same species. **Canis vulgaris**. How do we know that all dogs are members of the same species?

2. List some of the variation seen in the dog species.

Evolution

Evolution is a theory. Theory=A possible explanation for something
The Theory of Evolution=the characteristics of a species change over many generations.
Evolution has a few ideas in it:
All species are related.
Some species are more closely than others.
Different species may have evolved from a shared common ancestor.

Humans and Neanderthals do they share a common ancestor?

Up until about 30,000 years ago two species of humans existed together on the Earth. Homo sapiens (us) and Neanderthals. Some people believe these two

different species evolved from one common ancestor called Homo heidelbergensis.



Homo heidelbergensis

